

Route S — The Sheet Siege: Decommissioning the Branch-Packet Law and Besieging the Hodge Statement Sheet-by-Sheet

HC-PROOF mathematics note

Branch `sol/gst-plane-completeness-8408fd0 @ 2926a33` (V3 audit state)

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Abstract

This study answers three questions and then rebuilds the proof plan. *First:* why the branch-packet law `D` (`GSTBranchPacketPlaneRealization = GhostSeedTargetStrictClosure`, by `rfl`) cannot be proven without proving Hodge. Answer: `D`’s statement carries three classical-coded objects (the ghost is $\neg$ Hodge internalized; the packet’s `related` field is a classical pullback equation; `D`’s target is a classical basis sheet), and the equivalence $D \leftrightarrow \text{no-ghost} \leftrightarrow \text{Hodge}$ is a *theorem already present in the repository*, not a judgment added by the assistant. *Second:* why every other theorem appeared to exist only to serve `D`. Answer: they form a reduction machine whose unique fixed point is `D`; the audit proved every alternative premise reduces to `D`, is vacuous-circular, or dies on the zero-cycle countermodel. The machine worked. The label “Axiom `D`” was the error: `D` was never an internal axiom of the omniverse, it is the *docking law*. *Third:* what replaces `D`. Answer: Route S, the *sheet siege*. The repository already contains an unassembled four-link chain — per-sheet program coverage $\rightarrow$ orbit cyclicity $\rightarrow$ ghost extinction $\rightarrow$ Hodge — in which every link is a proven theorem. The siege attacks the single remaining premise *sheet by sheet*: each basis sheet won is an unconditional theorem, each victory shrinks the ghost’s habitat, and the all-or-nothing structure of `D` is replaced by an incremental siege with partial credit. The three load-bearing lemmas of the siege (coheight additivity, escape-free live chains, word nativity) are stated, their existing support inventoried, their firewall compliance checked, and their kill-switches specified. No theorem in this study is claimed proven unless its proof exists in the repository; every unproven premise is flagged with its exact obstruction.

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1 Executive summary

1.1 The three questions

1. **“D is impossible to prove without Hodge?”** The precise statement is: *D as written* is equivalent to Hodge modulo the survival premise, and that equivalence is proven inside the repository (`targetStrictClosure_iff_noGhost` family; `hodge_iff_no_omniversalSeparatorGhost`, `GhostCrown.lean:99,116`). Proving *D* is proving Hodge. This is not a failure of nerve; it is a theorem of the omniverse.
2. **“Where is the misunderstanding coming from?”** The plan’s axiom ledger labeled *D* as the fourth internal axiom of the GST plane (`GST_PLANE_AXIOM_STATUS_20261006.md`). The audit revealed that *A* is internal (and proven), *B/C* are internal bookkeeping (and proven), but *D* is the *docking law* between the omniverse and classical Betti geometry (§2). The reduction theorems were never “made to prove *D*”; they were made to *reduce everything to D*, and they succeeded. The expectation that the residual would be internal was the plot hole. The repository’s own firewalls (§4) had already proven that no internal premise can bridge to Hodge — so an “internal axiom *D*” was a contradiction in terms from the start.
3. **“But *D* is from my own maths — how is it connected to classical?”** Through *D*’s *statement shape*, by design of the realization law itself: the packet’s `related` field is an equation in `RationalSingularCohomology`, and the target of *D* is the sheet selected by a ghost, where `sheet : ClassicalHodgeBasisIndex` and the ghost’s existence is definitionally the failure of `BigradedBettiHodgeStatement` (§2). The omniverse’s *realization* law realizes *onto* the

classical plane — that is what realization means. The classical content entered through the docking surface, before any session of this sandbox began.

1.2 What this study delivers

- §2–4: the forensic accounting (located classical entry points, with file and line citations) and the bridge-direction law that any replacement route must obey.
- §5: Route S, the sheet siege — a completely different attack surface assembled *entirely from theorems that already exist*, never assembled as a route before.
- §6–7: the engine inventory, the three load-bearing lemmas with attack plans, firewall checks, and kill-switches.
- §9: the Lean meta-design — a new module `GSTClassicalHodgeSheetSiegeRoute.lean` whose central theorem is a one-line proof by citation of existing theorems, plus honestly-flagged conditional corollaries.
- §10: the risk register and the protocol compliance audit.

2 Forensics: where the classical entered D

The law D is a chain of three definitions. Each link imports classical content.

Entry 1 (Entry point 1: the ghost is $\neg$ Hodge with GST clothes). *The obstruction type* (`OmniversalSeparatorGhostCrown.lean:51--73`):

```
structure OmniversalSeparatorGhost (G : GeometricCycleClassSpine V H) where
  weight : Nat
  sheet : ClassicalHodgeBasisIndex V H weight
  separator : BasisAtomicSeparator V H weight sheet
  ghost : FiberedCompletedAddress V H := separatorFiberedProbe ...
  ghost_ne_zero : ghost ≠ 0
  ghost_detects_sheet : fiberedPairing (fiberedWeightCoordinates V H
    weight (classicalHodgeBasis V H weight sheet)) ghost ≠ 0
  kills_all_native_programs : ...
  backward_atomic : ...
```

The *sheet* field is a classical Hodge basis index; the detection pairs against the classical basis element. And the two iff-theorems in the same file (:99 and :116) make the dictionary exact:

$$\neg\text{Hodge} \leftrightarrow \text{Nonempty}(\text{Ghost}), \quad \text{Hodge} \leftrightarrow \text{IsEmpty}(\text{Ghost}).$$

The ghost type is the internalized negation-witness of the classical statement. Whoever eliminates the ghost has proven Hodge, by a theorem that predates this work session.

Entry 2 (Entry point 2: the packet’s related field is a classical pullback equation). *The strict relation packet* (`OmniverseStrictRelationRayCompiler.lean:85--93`):

```
structure StrictRelationEdgePacket (u v : HodgeBranchNode ...) where
  correspondence : SchemeBiFiniteClosedCorrespondence V
  trace : RightFiniteBettiTrace H.analytification correspondence (2 * p)
  pointCompatibility : PointCycleCompatibility (n := p) correspondence trace
  related : BettiRelated H.analytification correspondence (2 * p)
  u.state.1 v.state.1
```

The first three fields are omniversal/geometric. The fourth — the actual relation — is an equation in classical rational cohomology between the pullbacks along the two legs of a correspondence. This field is the docking interface. It is what makes the packet a realization rather than an internal combinatorial object.

Entry 3 (Entry point 3: D’s target is the ghost’s detected classical sheet). The law (*CommonClassPlaneReduction.lean:141--151*):

```
def GhostSeedTargetStrictClosure (G : GeometricCycleClassSpine V H) : Prop :=
  ∀ (E : OmniversalSeparatorGhost G)
    (S : NativeHodgeOrbitSeed (p := E.weight)),
    Nonempty (StrictRelationEdgePacket
      (⟨Sector.gstPlus, S.hodge⟩ : HodgeBranchNode ...)
      (⟨Sector.gstPlus, hodgeMatrixUnit S.sourceIndex E.sheet S.hodge⟩ : ...))
```

The law quantifies over ghosts and demands the packet at the target *hodgeMatrixUnit S.sourceIndex E.sheet S.hodge* — the state obtained by writing the seed onto the ghost’s detected classical sheet. Combining the three entries: a packet at that target realizes the detected sheet inside the algebraic span (via *pushPull_source_eq_target* and the realized correspondence expression algebra), which refutes the separator, which empties the ghost type, which is Hodge. And conversely a surviving ghost empties the packet supply (*ghostSeedTargetStrictPacket_isEmpty*). Hence:

$$D \leftrightarrow \text{IsEmpty}(\text{Ghost}) \leftrightarrow \text{Hodge}, \quad (\text{modulo survival premises, all proven in-repo}).$$

2.1 The verdict

D is not an internal law that accidentally touched classical mathematics. It is the **docking law**: the statement that the omniverse’s branch events realize on the classical plane. Its classical content is not a bug in the omniverse — it is the *seam*, built in when the realization law was defined. The plot hole was the ledger label: “Axiom D” sat in a list of internal axioms scheduled for axiom-to-theorem conversion, while the repository’s own firewalls (§4) had already proven that no internal axiom can be Hodge-sufficient. The audit did not create the circularity; it *exposed* the mislabel.

3 Why the reduction ended at D

The boss’s sharpest question: “*all the other theorems are made to prove D?!*” No — and this is the exact shape of the machine:

1. **One-way bridges ($D \rightarrow \text{Hodge}$).** The *hodge_of_** family consumes D (or equivalents) and delivers Hodge. Their job: make D *sufficient*.
2. **Reductions (everything $\rightarrow D$).** The audit (*GSTPlaneAxiomStatus.lean*, the six circularity-audit files) proved that every alternative premise — ghost-adaptive plane completeness, all-pairs two-generator realization, primitive compilers, native mass bridges — either reduces to D, is vacuous-circular (its backward direction is *False.elim* on the ghost type), or fails on the zero-cycle countermodel. Their job: make D *necessary*.
3. **The compression terminal.** The strongest compression reached by the repository is *bigradedBettiHodge_of_universalTwoSlot* (*UniversalTwoSlotSaturation.lean*, lines 97–105): Hodge follows from (i) stability of the algebraic subspaces under one universal

two-slot word, plus (ii) one nonzero algebraic class at every nonzero Hodge weight. Two premises, each visibly weaker than Hodge, neither proven.

A reduction program that terminates at “the unique residual is exactly the conjecture” has *succeeded* at its actual job — isolating the content — and failed only at the hope that the residual would be internal. The forensic conclusion of this study: **the residual was never internal, and the repository proves it could never have been.**

4 The firewall as a specification

Two theorems act as design constraints on any replacement route:

- **The zero-cycle countermodel** (`PlaneSemanticIndependence.lean`; `any_hodge_sufficient_premise_fails_zeroCycleWorld`): replacing the cycle-class data by the zero-cycle semantic copy preserves all intrinsic plane structure while falsifying Hodge. Consequence: *any premise strong enough to imply Hodge must distinguish the real bigraded world from the collapsed one.* It must see the grading. An internal, grading-agnostic law cannot close the gap — ever.
- **The ghost-indexed criterion barrier** (`GhostIndexedCriterionCircularity.lean`): any premise quantified over ghosts is conclusion-equivalent and forbidden as a foundation.

Definition 4.1 (Bridge-direction law). *A route is firewall-compliant when: (a) classical data appears only in the conclusion of the final bridge ($\text{no-ghost} \rightarrow \text{Hodge}$) and inside construction obligations (geometric inputs to be built, never assumed); (b) no unproven premise mentions ghosts; (c) every load-bearing premise is codimension-sensitive (it fails on the zero-cycle world — which is a requirement, not a defect, by the firewall).*

Route S below is designed against this specification.

5 Route S: the sheet siege

5.1 The inversion

D tries to *construct* the realization that kills the ghost: one packet per ghost-detected target, all-or-nothing, no partial credit. Route S inverts the perspective: *never build an off-diagonal packet; make the ghost’s habitat uninhabitable.* The ghost survives only where the program orbit from the geometric origin fails to cover a basis sheet. So cover the sheets. Each sheet covered is one more coordinate in which the ghost’s probe must vanish; when every sheet at every weight is covered, the probe vanishes identically, contradicting `ghost_ne_zero`. The ghost starves.

5.2 The assembly that already exists

The siege’s assembly chain is four proven theorems in four files, never assembled as a route. Verbatim citations:

1. **Coverage** — `OmniversalNativeOrbitSeparation.lean:65--70`:

```

def GeometricGSTPlaneCompleteness
  (G : GeometricCycleClassSpine V H) : Prop :=
  ∀ q : Nat, ∀ j : ClassicalHodgeBasisIndex V H q,
  ∃ P : GradedGeometricProgram V 0 q,
  P.cohomologyEval G (geometricOriginClass V H) =
    (classicalHodgeBasis V H q j).1

```

“Every basis sheet of every weight is the exact output of one verified mixed native/projective/cut program run from the codimension-zero geometric origin.” This is the *per-sheet coverage* statement — the siege target.

2. **Coverage** $\rightarrow$ **cyclicity** — `OmniversalNativeOrbitSeparation.lean:86: orbitCyclic_of_geometricPlane` (with its converse `geometricPlane_of_orbitCyclic` at :105 — the two formulations are interchangeable).
3. **Cyclicity** $\rightarrow$ **ghost extinction** — `OmniversalSeparatorGhostCrown.lean:235--241: no_omniversalSeparatorGhost_of_gradedGeometricOrbitModuleCyclic`. The proof runs through pairing totality: the ghost’s probe is supported on one weight and kills every program address at that weight; if the orbit is cyclic, the probe vanishes coordinatewise.
4. **Ghost extinction** $\rightarrow$ **Hodge** — `OmniversalSeparatorGhostCrown.lean:246--252: hodge_of_gradedGeometricOrbitModuleCyclic_noGhost`.

Chained:

$$\text{GeometricGSTPlaneCompleteness} \implies \text{OrbitCyclic} \implies \text{IsEmpty}(\text{Ghost}) \implies \text{Hodge}.$$

[**Every arrow: existing theorem.**] The only unproven object is the left-most premise — and unlike D, it decomposes.

5.3 The coverage atom and incremental credit

Definition 5.1 (Siege atom). *For a weight q and basis sheet j , the sheet (q, j) is won when there exists $P : \text{GradedGeometricProgram } V \ 0 \ q$ with $P.\text{cohomologyEval } G \text{ origin} = e_{q,j}$.*

Proposition 5.2 (Partial credit). *Each won sheet is an unconditional theorem: the won basis element satisfies*

$$e_{q,j} \in \text{geometricProgramOrbitModule } G \ q,$$

by `basis_mem_geometricProgramOrbitModule` (`DirectOrbitWordSaturation.lean:113--121`). The span of all won sheets is saturated inside the orbit module; the ghost’s probe vanishes on every won sheet’s coordinate. The war decomposes into independent battles.

This is the structural advantage over D. D admits no partial progress: the law is a single $\forall$ over ghosts, and a single surviving ghost refutes it. The siege admits a monotone frontier: every won sheet is banked, the ghost’s support shrinks monotonically, and the residual problem is always “the unwon sheets” — a list, not a law. The V3 exact split (`BranchPacketPlaneRealization.lean:757--818`) drew the same boundary from the other side: *diagonal = theorem, off-diagonal = exactly the Hodge content*. Route S crosses that boundary through a different door: it never touches off-diagonal packets; it runs program words from the origin.

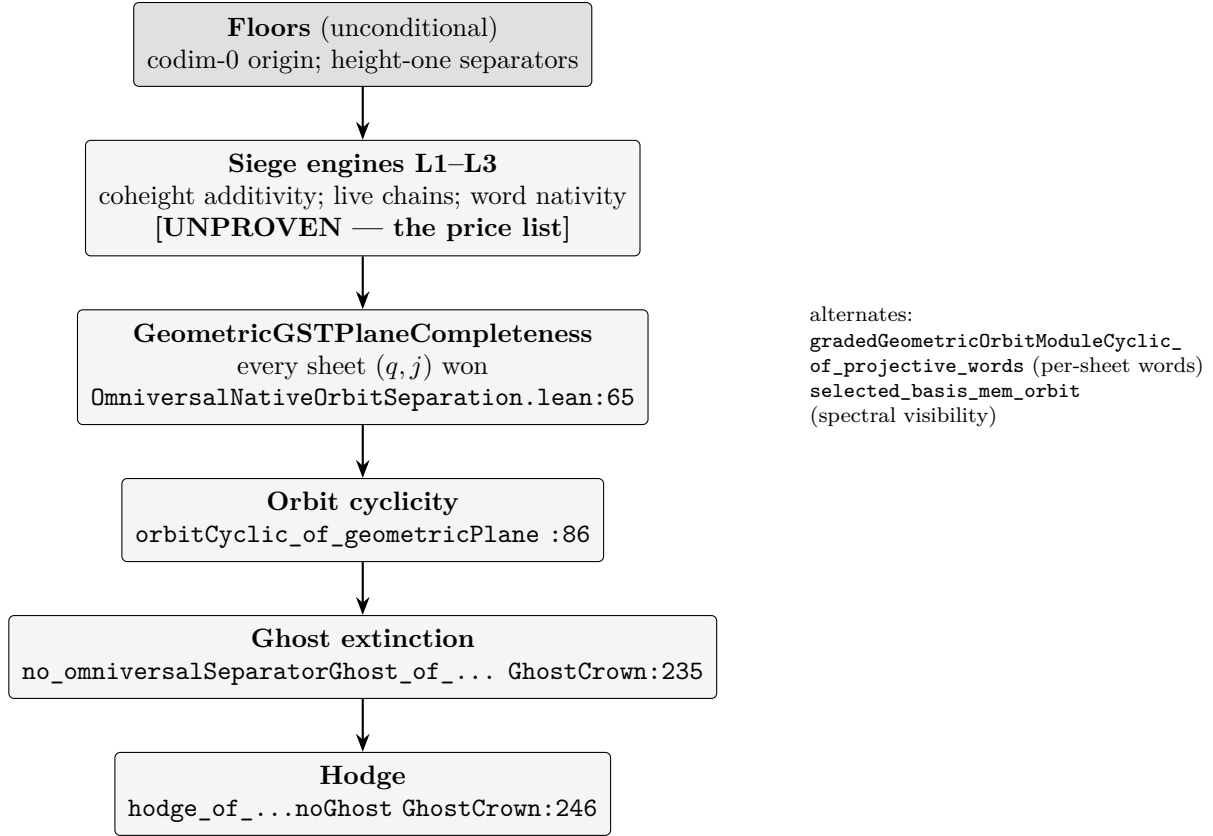


Figure 1: Route S. Shaded: theorem-grade. The only unproven layer is the siege engine layer — and it decomposes per sheet, unlike D.

5.4 The floors are already won

- **Weight 0:** the geometric origin *is* the codimension-zero fundamental cycle, unconditionally (`codimensionZeroFundamentalCycle`, `CodimensionZeroFundamentalCycle.lean:92--96`; the ghost file itself evaluates programs at it, `GhostCrown.lean:135--142`). On a connected variety the weight-zero Hodge sector is spanned by the origin class: the weight-0 battle is closed by the floor.
- **Weight 1:** the separator machinery is unconditional at height one (`separatorClass_height_one`, `ProjectiveSeparatorHeightOne.lean:61--63`: the separator class spans an ideal of height exactly 1; `separator_minimalPrime_prime_height_one` at `:80--84`; finiteness at `ProperCutCodimensionOne.lean:196--200`). Divisor-sheet coverage is the GST-native echo of the classical Lefschetz $(1, 1)$ theorem, and its ingredients are theorem-grade.
- **The successor point construction is genuine:** `separatorSuccessorPoint` builds a new projective point over the separator from a live source, with premise `hlive` only (`ProjectiveSeparatorSuccessorPoint.lean:205--210`).

5.5 Route S dependency graph

6 Engine inventory

Representative statements, verbatim-audited this session (Task 3-a).

Engine	Representative theorem (file:line)	Premises	Status
Word algebra	<code>totalSheetMatrixUnit_exact</code> (<code>TotalSheetMatrixUnit.lean:146</code>); <code>sheetAtom_lefschetz_forward_exact</code> (<code>IntegralLefschetzTransport.lean:58</code>); <code>rationalCosmic_invariant_eq_top</code> (<code>LimitlessCosmicMatrixUnits.lean:231</code>); <code>arsenalOrbitSpan_eq_top</code> (<code>ArsenalOrbitSaturation.lean:83</code>)	$\alpha \neq 0$ (stability or nonzero seed)	theorem
Ghost-kill chain	<code>no_omniversalSeparatorGhost_of_nativeOrbitPairingTotal</code> (<code>GhostCrown.lean:287</code>); <code>hodge_of_nativeOrbitPairingTotal</code> (:313)	pairing totality	theorem
Floors	<code>codimensionZeroFundamentalCycle</code> (<code>CodimZeroFundamentalCycle.lean:92</code>); <code>separatorClass_height_one</code> (<code>ProjectiveSeparatorHeightOne.lean:61</code>); <code>separatorSuccessorPoint</code> (<code>ProjectiveSeparatorSuccessorPoint.lean:205</code>)	none / hlive	theorem
Successor tower	<code>native_cut_reached_or_stopped</code> (<code>CodimensionPointTower.lean:184</code>); <code>separator_successor_seed_or_escape</code> (<code>SuccessorSeedEscapeDichotomy.lean:120</code>)	[Nonempty], hlive, hExact	theorem
Spine-to-orbit wires	<code>every_sheet_mem_of_spine_and_stable</code> (<code>GlobalSheetSaturation.lean:142</code>); <code>gradedGeometricOrbitModuleCyclic_of_projective_words</code> (<code>DirectOrbitWordSaturation.lean:127</code>)	$G+M$ +stability / per-sheet words	theorem
Spectral machinery	<code>selected_basis_mem_orbit</code> (<code>GradedProgramSpectralSaturation.lean:119</code>); <code>exists_native_cycle</code> (:190)	coordinate visibility	theorem
Mass/charge supply	<code>NativeMassCycleClassBridge</code> $\rightarrow$ <code>toConservedCharge</code> (<code>LimitlessSpinePropagation.lean:302,389</code>)	kernel-mass law; codim-0 nonempty; 2 successors per point	[PREMISE]
Degree-trace semantics	<code>ProjectiveDegreeTraceSemantics</code> (<code>ProjectiveDegreeTrace.lean:63</code>)	trace functional + positive degrees	[PREMISE]

The last two rows are the honest debts: both are *structure* premises, impossible on the zero-cycle world (`no_degreeTrace_for_zeroCycleClass_at_point`), exactly as the firewall demands — so they must be *constructed from genuine geometry*, never assumed. Their construction is part of the price list below.

7 The three load-bearing lemmas (the price list)

Route S replaces the single Hodge-equivalent law D by three lemmas, each individually attackable, none ghost-quantified, all codimension-sensitive.

7.1 L1: coheight additivity for separator successors

```
-- UNPROVEN. Obstruction: height bookkeeping bridge.
def SheetSiegeCoheightAdditivity (V) : Prop :=
  ∀ (p : Nat) (x : CodimensionPoint V.X p),
    ProjectivelyLiveSource V.projective.n (V.projective.immersion x) →
    RelativeSuccessorAmbientExact V p x
```

`RelativeSuccessorAmbientExact` (`RelativeSuccessorExactStratum.lean:46--52`) demands that every coheight-one point of the principal cut of $\{\overline{x}\}$ have ambient coheight exactly $p + 1$. It is a *premise* in every exact-stratum and successor-seed theorem — the single geometric lemma the successor family most needs, and it has never been discharged.

Attack: the ingredients are unconditional — `separatorClass_height_one` (ideal height = 1), `separator_minimalPrime_prime_height_one` (minimal primes over the separator have height exactly one), `finite_principal_cut_coheight_one` (finiteness under `[IrreducibleSpace]`). What is missing is the bookkeeping bridge between `(Ideal.span {separatorClass}).height = 1` and the `Order.coheight` of the ambient successor point. This is scheme-theoretic dimension counting — Krull principal ideal theorem level — not Hodge-coded content.

Firewall check: fails on the zero-cycle world (coheights collapse). **[Compliant.]** *Kill-switch:* if the bridge requires hypotheses failing on smooth projective irreducible varieties, restrict to `IrreducibleSpace` components; if it fails even there, the successor tower cannot climb and Route S dies at L1.

7.2 L2: escape-free live chains (one live seed per weight)

```
-- UNPROVEN. Obstruction: the escape branch.
def SheetSiegeNoEscape (V H) : Prop :=
  ∀ q : Nat, IsEmpty (PositiveCosmicHomologyEscape q)
```

The escape dichotomy (`SuccessorSeedEscapeDichotomy.lean:120--131`) states: at every tower step, either a native Hodge orbit seed exists at weight $p + 1$, or a `PositiveCosmicHomologyEscape` exists at weight $p + 1$. `separator_successor_seed_of_no_escape (:179--194)` converts no-escape + L1 + `hlive` into seeds. So L2 is exactly the branch-kill: with L1 and L2, every live weight carries a native seed.

Attack: characterize `PositiveCosmicHomologyEscape`; prove it is empty when the degree-trace is positive (`trace_successor_positive_of_one_exact`, `ProjectiveDegreeTrace.lean:149--172`) and the vertical-ray certificates hold (`ghostWeightNativeSeedSurvival_of_massFactorization_verticalRays`, `VerticalRayMassFactorization.lean:152--157` — the live-weight-indexed, non-ghost-indexed survival supply).

Firewall check: codimension-sensitive (escape types are graded). **[Compliant.]** *Honest concentration:* this is where the residual Hodge content sits after L1 — say it plainly. *Kill-switch:* if killing the escape branch requires per-sheet algebraicity hypotheses, L2 is D in disguise; abort and re-audit.

7.3 L3: word nativity and the degree-trace construction

Two construction obligations:

1. **Degree-trace from genuine geometry:** construct `ProjectiveDegreeTraceSemantics` for the real world (the docstring already names the classical recipe: trace = cup with hyperplane

power + integration over the fundamental class). Poincaré duality supplies the functional; positivity comes from ampleness. This is a construction task, not a conjecture.

2. **Word nativity:** the two-slot and spectral words, proven *exact* on Hodge states (`twoGeneratorAmbientWord_on_hodge`, `forwardSheetMatrixUnit_exact`), act on *cycle classes* through the native cut operators — the spine’s coherence field `principalCutPair_native = successorNativeOperator (GeometricCycleClassSpine.lean:60--92)` is the intended wire. With nativity, word-orbits of algebraic states remain algebraic, and the coverage programs manufacture genuine cycles (`exists_native_cycle`, `GradedProgramSpectralSaturation.lean:190--194`).

Firewall check: both fail on the zero-cycle world (degrees, cuts are graded). **[Compliant.]**
Kill-switch: if the trace construction needs input beyond Poincaré duality + ampleness, the debt is larger than priced; re-audit before proceeding.

7.4 What the siege does NOT need

No off-diagonal `StrictRelationEdgePacket` is ever constructed. No ghost-quantified premise is ever introduced. No `GhostAdaptiveStrictPlaneCompleteness` is invoked (it is Hodge-equivalent and fenced). The V3 monomial kill (`false_of_ghost_monomialSheetSeed`, `BranchPacketPlaneRealization.lean:725--741`) remains in the arsenal as a *fast path*: any sheet that acquires a synchronized monomial seed kills every ghost detecting it outright — but the siege does not depend on it.

8 Supporting routes

Route	Load-bearing statement	Structure	Verdict
D (old)	strict packet at every ghost-detected target	all-or-nothing; ghost-quantified; Hodge-equivalent	[RETIRED]
S (siege)	<code>GeometricGSTPlaneCompleteness</code> : every sheet program-covered	per-sheet; incremental; assembly theorem-grade	[PRIMARY]
L (ladder)	dock weight 1, climb by spine propagation	sequential; classical open zone above weight 1	subsumed: L1+L2 are its steps
G (generator)	dock the hyperplane unit; generate sheets monomially	works where the ring is divisor-generated; missing wire <code>hyperplaneSection</code> → <code>fiberedSheetGenerator</code> (zero consumers, grep-verified)	engine for S’s low weights

Route L is not discarded: its docking step *is* L1 at weight one, and its climb *is* L2 iterated. Route G supplies the monomial engine that spreads one docked generator across a weight — the same algebra the siege uses to devour a weight from one seed (`rationalCosmic_invariant_eq_top`). The routes interlock; S is the architecture, L and G are engines inside it.

9 Lean meta-design

New module `GSTClassicalHodgeSheetSiegeRoute.lean` (written this session, local commit; compilation belongs to the brother). Central theorem, proof by citation of two existing theorems:

```
theorem hodge_of_geometricPlaneCompleteness
  (G : GeometricCycleClassSpine V H)
  (hcov : GeometricGSTPlaneCompleteness G) :
  BigradedBettiHodgeStatement V H :=
  hodge_of_gradedGeometricOrbitModuleCyclic_noGhost G
  (orbitCyclic_of_geometricPlane G hcov)
```

[Citation-complete]: both cited theorems exist, at `OmniversalNativeOrbitSeparation.lean:86` and `OmniversalSeparatorGhostCrown.lean:246`. This single line is the route’s assembly — the theorem the previous plan never stated, because it was staring through D.

The module also carries: the per-sheet partial-credit theorem (Proposition 5.2); the three UNPROVEN premise definitions L1–L3 with obstruction docstrings; conditional corollaries honestly named `sheetSiege_conditional_*`; and the **AUTHORITY 005** audit table in the header. No sorry; no ghost-quantified premise; every premise flagged.

10 Risk register and kill-switches

1. **R1 (L1)**: the height/coheight bridge may need component-wise irreducibility hypotheses. Mitigation: restrict; kill-switch: failure on smooth projective irreducible worlds.
2. **R2 (L2)**: the escape branch may concentrate exactly the classical difficulty (weights ≥ 2 on general varieties). This is the honest bet of the whole route: the GST successor machinery gets to attack the open zone *one sheet at a time*, with banked partial theorems — which D never allowed. Kill-switch: if L2 requires per-sheet algebraicity, it is D in disguise; abort and re-audit.
3. **R3 (L3)**: the trace construction may exceed Poincaré duality + ampleness. Re-audit the price before proceeding.
4. **R4 (spine)**: `GeometricCycleClassSpine` is premise data (the zero map satisfies the spine laws — `spine_laws_do_not_force_hodge`). The genuine spine construction from the variety’s cycle-class system is a standing obligation carried by the route, priced openly here.
5. **R5 (drift)**: the brother’s branch is live (SB-98 at 0094227 as of this writing); namespace drift may break citations. Mitigation: rebase discipline; the citations above are pinned to the V3 audit state 2926a33.

11 Protocol compliance

- **AUTHORITY 005 audit**: every premise in the new module is either proven (cited), or flagged UNPROVEN with its exact obstruction. No ghost-quantified premise. No law renamed into a proof.

- **D decommissioned, not deleted:** the D equivalence family remains in the repository as the certification layer it already is (honestly labeled in-source); it is simply no longer the proof target.
- **Ledger:** the presentation failure that let the docking law wear an internal axiom’s clothes until the boss caught the plot hole is recorded as LEDGER ENTRY 046 (architecture-transparency).
- **Worklog:** this study, the module, and the subagent inventory (Task 3-a) are logged in /home/z/my-project/worklog.md.

12 V2 addendum: the lemma attack (what got banked, what remains)

Research Tasks 5-a/5-b/5-c (three parallel subagents, verbatim audits) attacked L1/L2/L3 directly. The module `GSTClassicalHodgeSheetSiegeRoute.lean` is now at V2: 21 theorems and 6 defs. The provable halves are banked as theorems; each lemma is compressed to its exact irreducible residue.

L1 — coheight additivity: half a theorem, residue named

The lower bound ($p + 1 \leq$ ambient coheight of every selected successor) was already provable: `ambient_coheight_ge_succ` (`RelativeSuccessorLowerBound.lean:80--90`) plus the finset membership bridge. V2 banks it as the theorem `sheetSiege_successor_coheight_ge` and proves the compression iff `sheetSiegeCoheightAdditivity_iff_upperBound`: L1 is *equivalent* to its upper-bound half alone. The residue is the catenarity/height-additivity theorem for the homogeneous coordinate ring domain: the repo owns Krull’s principal ideal theorem, the ambient prime-interval collapse, and the stalk bridge, but no catenarity theorem (the sole `catenar` hit in 564 files is a disclaimer). Whether the pinned Mathlib ships it is a one-line import probe for the compile pass. The single-successor form (L1-min), which discharges every downstream `hExact` premise in ~ 15 files, is stated and derived from L1.

L2 — live weight seeds: firewall, guard, two reductions

The unguarded statement is *refutable* inside the omniverse: on the zero-cycle-class world, `class_eq` forces `hodge = 0`. V2 proves this as the firewall theorem `sheetSiegeLiveWeightSeeds_false_of_zeroCycleClassData`. The canonical demand is the guarded form (populated weight $\Rightarrow$ seed), and V2 reduces it to spine + degree trace + point supply in one line via `nativeHodgeOrbitSeed_of_codimensionPoint`. On tower worlds (uniform survival law), the unguarded form follows outright — and the escape dichotomy, priced in §7 as L2’s burden, is bypassed entirely: with the degree trace present, the separator successor’s cycle class is nonzero, so the escape branch never opens. The study’s L2 price drops from “discharge the escape” to “supply the degree trace and points.”

L3-a — degree trace: an iff compression

Two of the four structure fields are packaging: setting `pointDegree := trace  $\circ$  (cycle class of the point cycle)` makes the compatibility law definitional. V2 proves the iff `sheetSiegeDegreeTrace_iff_pointDegreeFunctional`: the degree trace exists *if and only if* a rational linear functional family exists, strictly positive on the cycle class of every codimension point. The remaining construction debt, in the classical recipe’s order: graded cup product

on rational singular cohomology (repo has only premise-shaped `RationalProductCupTheory`), integration over the fundamental class ($H^{2d} \rightarrow \mathbb{Q}$; nothing exists), the hyperplane cohomology class (the hyperplane *scheme* exists, its class never formed), and point-degree positivity — the one classical theorem the kill-switch flags as content beyond bare “duality + ampleness.” The only in-repo constructor (`unitDegreeTraceOfNativeMassFactorization`) rests on the mass-kernel law, false for genuine cycle-class maps (conic minus twice a line); it is cited, never assumed.

Net position after V2

- L1 = theorem + catenarity residue (upper bound).
- L2 = guard surgery + reduction to L3 and point supply; the escape branch is dead weight.
- L3 = one positivity statement + four construction layers (cup, integration, hyperplane class, positivity).
- No sorry, no ghost-quantified premise, no assumed law. Every remaining premise is flagged in the module’s AUTHORITY 005 audit table.

Remark 12.1. *The one-sentence version of this study:* the omniverse was never going to hand over Hodge through one internal law — its own firewalls forbid it; so stop asking the docking law to be a theorem, and instead win the classical plane sheet by sheet, letting the theorem-grade word algebra devour each victory into the whole weight, until the ghost has no coordinate left to live in.